

Percentage → Decimal → Fraction: Division Cheatsheet

For GATE aptitude, probability, algorithms and mathematical calculations

How to convert a percentage into a fraction

Example: $12.5\% = 12.5/100 = 125/1000 = 25/200 = 5/40 = 1/8$. At each step, divide the numerator and denominator by the same number until no common factor remains.

Percentage	Decimal	Full division / simplification	Final fraction
10%	0.1	$10/100 = 1/10$	$1/10$
12.5%	0.125	$12.5/100 = 125/1000 = 25/200 = 5/40 = 1/8$	$1/8$
16.66...%	0.1666...	$16.66.../100 = 1/6$ (recurring)	$1/6$
20%	0.2	$20/100 = 2/10 = 1/5$	$1/5$
25%	0.25	$25/100 = 1/4$	$1/4$
30%	0.3	$30/100 = 3/10$	$3/10$
33.33...%	0.3333...	$33.33.../100 = 1/3$ (recurring)	$1/3$
40%	0.4	$40/100 = 4/10 = 2/5$	$2/5$
50%	0.5	$50/100 = 5/10 = 1/2$	$1/2$
60%	0.6	$60/100 = 6/10 = 3/5$	$3/5$
62.5%	0.625	$62.5/100 = 625/1000 = 125/200 = 25/40 = 5/8$	$5/8$
66.66...%	0.6666...	$66.66.../100 = 2/3$ (recurring)	$2/3$
75%	0.75	$75/100 = 3/4$	$3/4$
80%	0.8	$80/100 = 8/10 = 4/5$	$4/5$
83.33...%	0.8333...	$83.33.../100 = 5/6$ (recurring)	$5/6$
87.5%	0.875	$87.5/100 = 875/1000 = 175/200 = 35/40 = 7/8$	$7/8$
90%	0.9	$90/100 = 9/10$	$9/10$
100%	1	$100/100 = 1$	1

Quick mental shortcuts

Fraction	Percentage	Decimal
$1/2$	50%	0.5
$1/3$	33.33...%	0.3333...
$1/4$	25%	0.25
$1/5$	20%	0.2
$1/6$	16.66...%	0.1666...
$1/7$	14.2857...%	0.142857...
$1/8$	12.5%	0.125
$1/9$	11.11...%	0.1111...
$1/10$	10%	0.1

GATE-style Practice Questions

These are practice questions modeled on common GATE aptitude/math patterns, not verbatim PYQs.

1. A value is increased by 25% and then decreased by 20%. What is the net percentage change?

Answer: Increase: $\times 1.25$; decrease: $\times 0.80$; net factor = $1.00 \rightarrow 0\%$.

2. A machine completes $\frac{3}{8}$ of a task in 12 minutes. How long will it take to complete the whole task at the same rate?

Answer: $12 \div (\frac{3}{8}) = 12 \times \frac{8}{3} = 32$ minutes.

3. 12.5% of a number is 45. Find the number.

Answer: $12.5\% = \frac{1}{8}$, so number = $45 \times 8 = 360$.

4. A probability is 0.1666... Express it as a fraction and percentage.

Answer: $\frac{1}{6}$ and 16.66...%.

5. A student answers 60% of 50 questions correctly. How many are incorrect?

Answer: Correct = 30; incorrect = 20.

6. If A is 40% of B, then A:B is what?

Answer: $40:100 = 2:5$.

7. A quantity changes from 80 to 100. What is the percentage increase?

Answer: $20/80 \times 100 = 25\%$.

8. What is 62.5% of 96?

Answer: $62.5\% = \frac{5}{8}$; $96 \times \frac{5}{8} = 60$.

9. A number is reduced by 12.5%. By what fraction is it reduced?

Answer: $12.5\% = \frac{1}{8}$.

10. If $\frac{2}{3}$ of a job is completed, what percentage remains?

Answer: $\frac{1}{3}$ remains = 33.33...%.

11. A data packet has 640 bits. If 87.5% are payload bits, how many payload bits are there?

Answer: $87.5\% = \frac{7}{8}$; $640 \times \frac{7}{8} = 560$ bits.

12. A value is 20% less than another value. The smaller value is what fraction of the larger?

Answer: $80\% = \frac{4}{5}$.